

Python



Easy to learn & high-level programming

Language

Applications of Python Programming

↳ Developing websites → Frameworks like Django &

Flask

↳ Database connectivity → MySQL, PostgreSQL,

SQLite

↳ Data handling & manipulation → Pandas,
Numpy,

Matplotlib & seaborn

↓
Data science → ML, DL, NLP

↳ Automation & Scripting

How Python is different from other programming languages?

- ① Python's code simplicity
(simple & readable syntax)

- ## ② Dynamic typing

```
int Q = 10
```

Create an array → C++, Java

define the size of
an array prior

$$a = 24$$

List

24, 25.6, "Priya", True

Python, We don't need to define any
size prior

- ### ③ Memory Management

new-list

0	40
1	55
2	75
3	80
4	25
5	62
6	73
7	
8	
9	

$$10 \times 2 = 20$$

is very
slow

[illegible]

Features of Python

① Interpreted Language

↳ code is executed line by line.

② Cross-platform Language

↳ Python code on various Operating system

③ Integrated

↳ C, C++ → Python works very

④ Large standard library smoothly

↳ Numpy,

Panda,

Matplotlib,

Seaborn

these

programming

languages

How Python is used in data science?

① Data ingestion → Pull in data

(Data collection)

from APIs,
SQL queries

{ 50000, 70000,

90000, 75000,

5000, 1 cr }

② Data cleaning

→ clean, reshape, merge

Handling missing
Value, outliers

Transformation with Numpy/Pandas

③ EDA → Visuals with
matplotlib or seaborn
Exploratory Data Analysis

④ Model training → sklearn in
ML model
implementation

⑤ Model Evaluation & Interpretation

```
print("Hello", end=" ")  
print("World!")
```

Hello World!

by default, we have
end = "\n"
↳ next line

```
print("Hello")  
print("World!")
```

Hello
World!

Variables in programming are like containers that hold data. They're used to store information that can be accessed and manipulated throughout the program..

① Store the data

② Manipulating data

③ Reusability

④ Readability

```
user_input = input("Enter your name:")  
print("My name is:", user_input)
```

user input

Priya Bharia

display as it is to the
console

Basic rules to define variable

Variable names can contain letters (a-z, A-Z), digits (0-9), and underscores (_). ———— ①

Variable names must start with a letter or an underscore. They cannot start with a digit or any special characters. ———— ②

Variable names are case-sensitive, meaning uppercase and lowercase letters are considered different. For example, the variable names Temp and temp are different. ———— ③

Python keywords (reserved words) cannot be used as variable names. For example

print = "Hello"

Variable

In this example, print is a Python keyword used for printing output to the console. When we try to use it as a variable name and assign the value "Hello" to it, we'll encounter a syntax error because print cannot be used as a variable name. ———— ④

Variable names cannot contain spaces or special characters, except for underscores (_). ———— ⑤

automatically
understand

dataRecord = 20

int

the

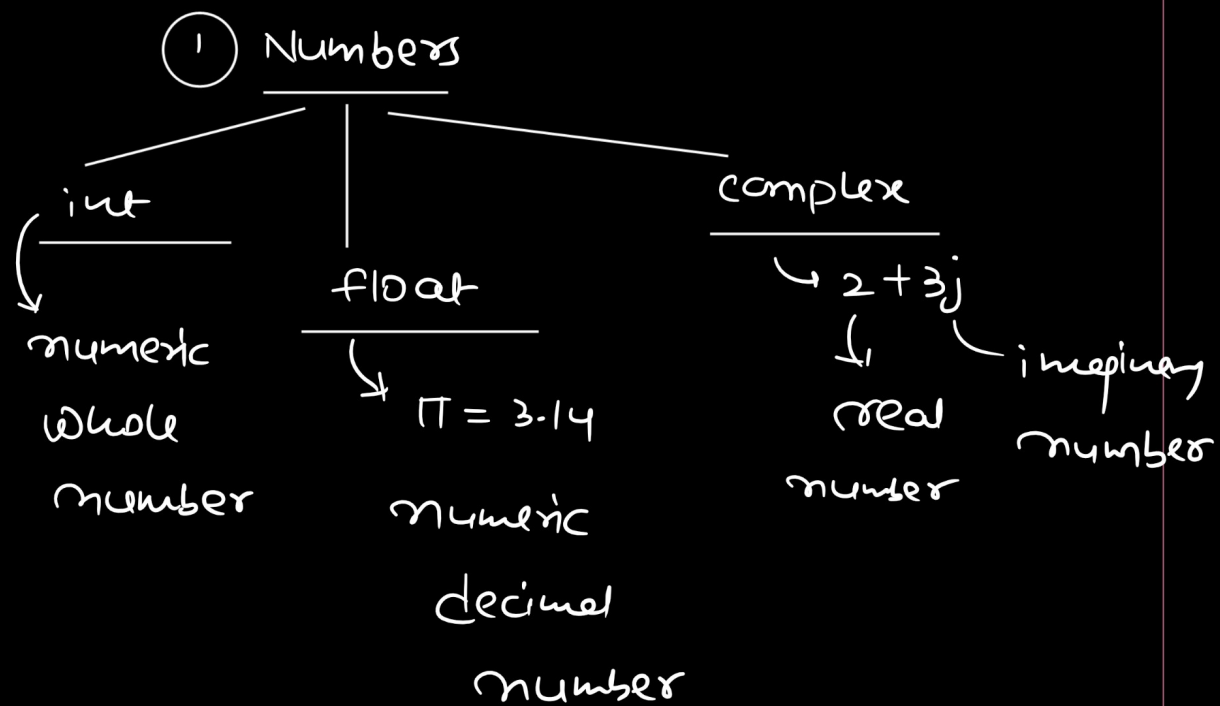
= → equal to (assignment operator)

datatype ≠

dataRecord

= = → equal equal to (comparison operator)

A data type tells a computer what kind of information you're working with. It helps the computer understand how to handle that information. For example, a data type could be "number," "text," "true/false," or "list of items."



② String → ' ' or " "